INSTRUMENT CLUSTER

DESCRIPTION

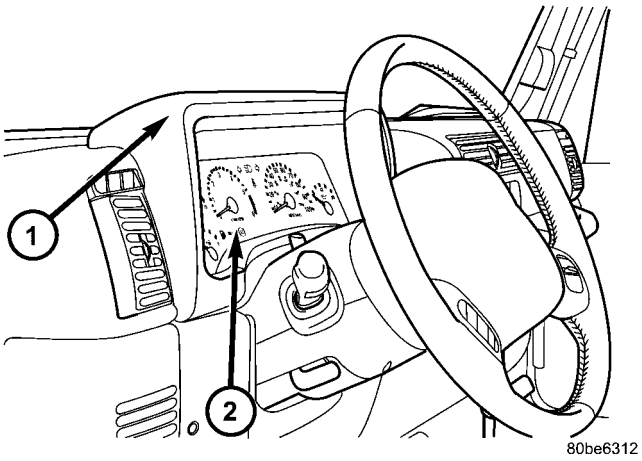

Fig. 1 Instrument Cluster

1 - CLUSTER BEZEL
2 - INSTRUMENT CLUSTER

The instrument cluster for this model is an ElectroMechanical Instrument Cluster (EMIC) that is located in the instrument panel above the steering column opening, directly in front of the driver (Fig. 1). The remainder of the EMIC, including the mounts and the electrical connections, are concealed within the instrument panel behind the cluster bezel. Besides analog gauges and indicators, the EMIC module incorporates a blue-green digital Vacuum Fluorescent Display (VFD) unit for displaying odometer/trip odometer information and certain diagnostic information. The instrument cluster for this model also includes the hardware and software necessary to serve as an electronic body control module.

The EMIC gauges and indicators are visible through a dedicated opening in the cluster bezel on the instrument panel and are protected by a clear plastic cluster lens (Fig. 2). Eight integral latch formations around the outer perimeter of the lens secure the lens unit to the cluster hood and the cluster housing. On the lower edge of the cluster lens just right of center, a small molded rubber boot covers the odometer/trip odometer switch button and seals a dedicated pocketed hole in the lens through which the switch button protrudes. Just behind the cluster lens is the cluster hood and an integral cluster mask, which are constructed of molded black plastic. The cluster hood serves as a visor and shields the face of the cluster from ambient light and reflections to reduce glare, while the cluster mask trims the outside perimeter of the cluster overlay. The cluster hood and mask unit is secured to the cluster

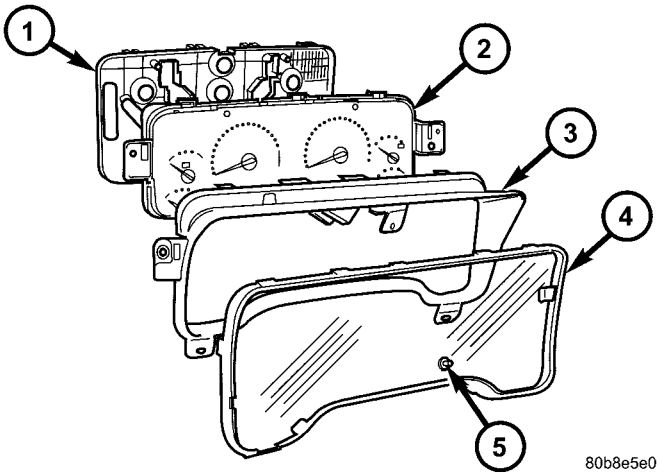

Fig. 2 Instrument Cluster Components

1 - REAR COVER
2 - CLUSTER HOUSING
3 - CLUSTER HOOD & MASK
4 - CLUSTER LENS
5 - SWITCH BUTTON BOOT

housing with five integral latch formations around its perimeter.

The rear of the cluster housing and the EMIC electronic circuitry are protected by a molded plastic rear cover, which is secured to the cluster housing with two screws, eight integral latch formations around its perimeter, and a single latch that is integral to the cluster housing. The rear cover includes clearance holes for service access to each of the five general illumination lighting and the one high beam indicator incandescent bulb and bulb holder units installed on the cluster circuit board, and for the two color-coded cluster connector receptacles. The connector receptacles on the back of the cluster electronic circuit board connect the EMIC to the vehicle electrical system through two take outs with color-coded connectors from the instrument panel wire harness.

Sandwiched between the rear cover and the lens, hood and mask unit is the cluster housing. The molded plastic cluster housing serves as the carrier for the cluster circuit board and circuitry, the cluster connector receptacles, the gauges, a Light Emitting Diode (LED) for each cluster indicator, the VFD unit, an audible tone generator, the cluster overlay, the gauge pointers, the odometer/trip odometer switch and the switch button. The molded plastic EMIC housing has four integral mounting tabs, two on the lower outboard edges of the housing and one on each side. A screw through each of these four mounting tabs secures the EMIC to the instrument panel structural support.

The cluster overlay is a laminated plastic unit. The dark, visible, outer surface of the overlay is marked with all of the gauge dial faces and graduations, but